Proje ödevi 4

1) Kodu :

```
%Three different methodes to Gauss elemenation
A = [ 2, -1, 2, 0;...
      2, -1, -1, 1;...
      2, 2, 1, 0;...
      2,-1,2,2];
b = [8;5;4;10];
%%
% first method
x_first = linsolve(A,b)
%%
% second method
x_second = A\b
%%
% third method
x_third = inv(A)*b
```

Çıktısı bu şekildedir:

```
x_first =

    2.2222
   -0.8889
    1.3333
    1.0000


x_second =

    2.2222
   -0.8889
    1.3333
    1.0000


x_third =

    2.2222
   -0.8889
    1.3333
    1.0000
```

2)Kodu:

```
A = [2, 3, -1;...
     1, -2, 2;...
     -1, 4, 1;];
b = [4;6;5];
fprintf("Determinant of A = %f\n",det(A) )

A1 = [4, 3, -1;...
      6, -2, 2;...
      5, 4, 1;];
fprintf("Determinant of A1 = %f\n",det(A1) )
fprintf("x1 = %f\n",det(A1)/det(A) )
A2 = [2, 4, -1;...
     1, 6, 2;...
     -1, 5, 1;];
```

```
fprintf("Determinant of A2 = %f\n",det(A2) )
fprintf("x2 = %f\n",det(A2)/det(A) )
A3 = [2, 3, 4;...
     1, -2, 6;...
     -1, 4, 5;];
fprintf("Determinant of A3 = %f\n",det(A3) )
fprintf("x3= %f\n",det(A3)/det(A) )
```

Çıktısı bu şekildedir :

```
>> question2
Determinant of A = -31.000000
Determinant of A1 = -62.000000
x1 = 2.000000
Determinant of A2 = -31.000000
x2 = 1.000000
Determinant of A3 = -93.000000
x3= 3.000000
>>
```

3)kodu:

```
A = [2, -1, 1;...
     3, 3, 9;...
     3, 3, 5;];
[L,U] = lu(A)
b=[-1;0;4];
A\b
L1 = [1,0,0;...
      3/2,1,0;...
      3/2,1,1];
U1 = [2,-1,1;0,9/2,15/2;0,0,-4];
```

Matlabta bulunan L ve U matrisleri:

L =

0.6667	1.0000	0
1.0000	0	0
1.0000	0	1.0000

U =

3	3	9
0	-3	-5
0	0	-4

Benim elle bulduklarım ise şöyledir:

L1 =

1.0000	0	0
1.5000	1.0000	0
1.5000	1.0000	1.0000

U1 =

```
   2.0000   -1.0000    1.0000
      0    4.5000    7.5000
      0         0   -4.0000
```

Benim bulduklarımın çarpımı de A yı vermektedir, böylelikle doğruluğu kontrol edildi.
x1,x2,x3 = 1,2 ,-1 şeklindedir.
4) Kodu :

```
A = [4,2,2;2,6,2;2,2,5];
A2 = [4,2;2,6];
fprintf("Determinant of A2=%f\n",det(A2))
A3 = [4,2,2;2,6,2;2,2,5];
fprintf("Determinant of A3=%f\n",det(A3))
R = chol(A);
R
Rt = transpose(R);
b = [4;6;-1];
index = [1;2;3];
fprintf("Solution with linsolve x%i=%f\n",[index,linsolve(A,b)].')
fprintf("Solution with A\\b x%i=%f\n",[index,(A\b)].')
fprintf("Solution with inv(A)*b x%i=%f\n",[index,inv(A)*b].')
```

Çıktısı ise şu şekildedir:

```
>> question4
Determinant of A2=20.000000
Determinant of A3=76.000000

R =

   2.0000    1.0000    1.0000
      0    2.2361    0.4472
      0         0    1.9494

Solution with linsolve x1=1.000000
Solution with linsolve x2=1.000000
Solution with linsolve x3=-1.000000
Solution with A\b x1=1.000000
Solution with A\b x2=1.000000
Solution with A\b x3=-1.000000
Solution with inv(A)*b x1=1.000000
Solution with inv(A)*b x2=1.000000
Solution with inv(A)*b x3=-1.000000
```

5)Kodu:

```
A = [1,2;-1,2];
[V,D] = eig(A)
A1 = [-1,2,0;0,3,4;0,0,7];
[V,D] = eig(A1)
```

Çıktısı ise bu şekildedir :

```
A =

     1     2
    -1     2


V =

   0.8165 + 0.0000i   0.8165 + 0.0000i
   0.2041 + 0.5401i   0.2041 - 0.5401i


D =

   1.5000 + 1.3229i   0.0000 + 0.0000i
   0.0000 + 0.0000i   1.5000 - 1.3229i


A1 =

    -1     2     0
     0     3     4
     0     0     7


V =

    1.0000    0.4472    0.1741
         0    0.8944    0.6963
         0         0    0.6963


D =

    -1     0     0
     0     3     0
     0     0     7
```